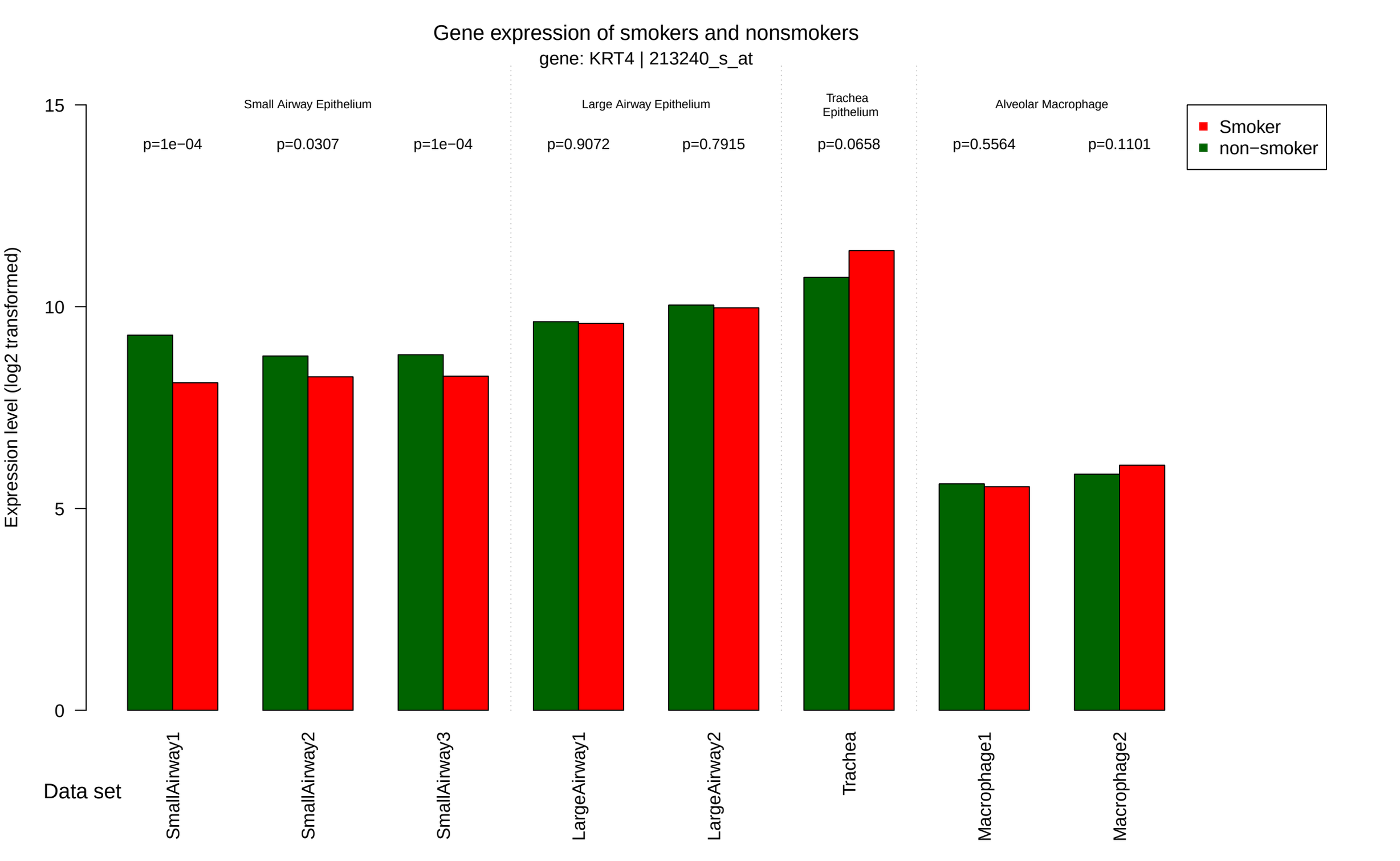
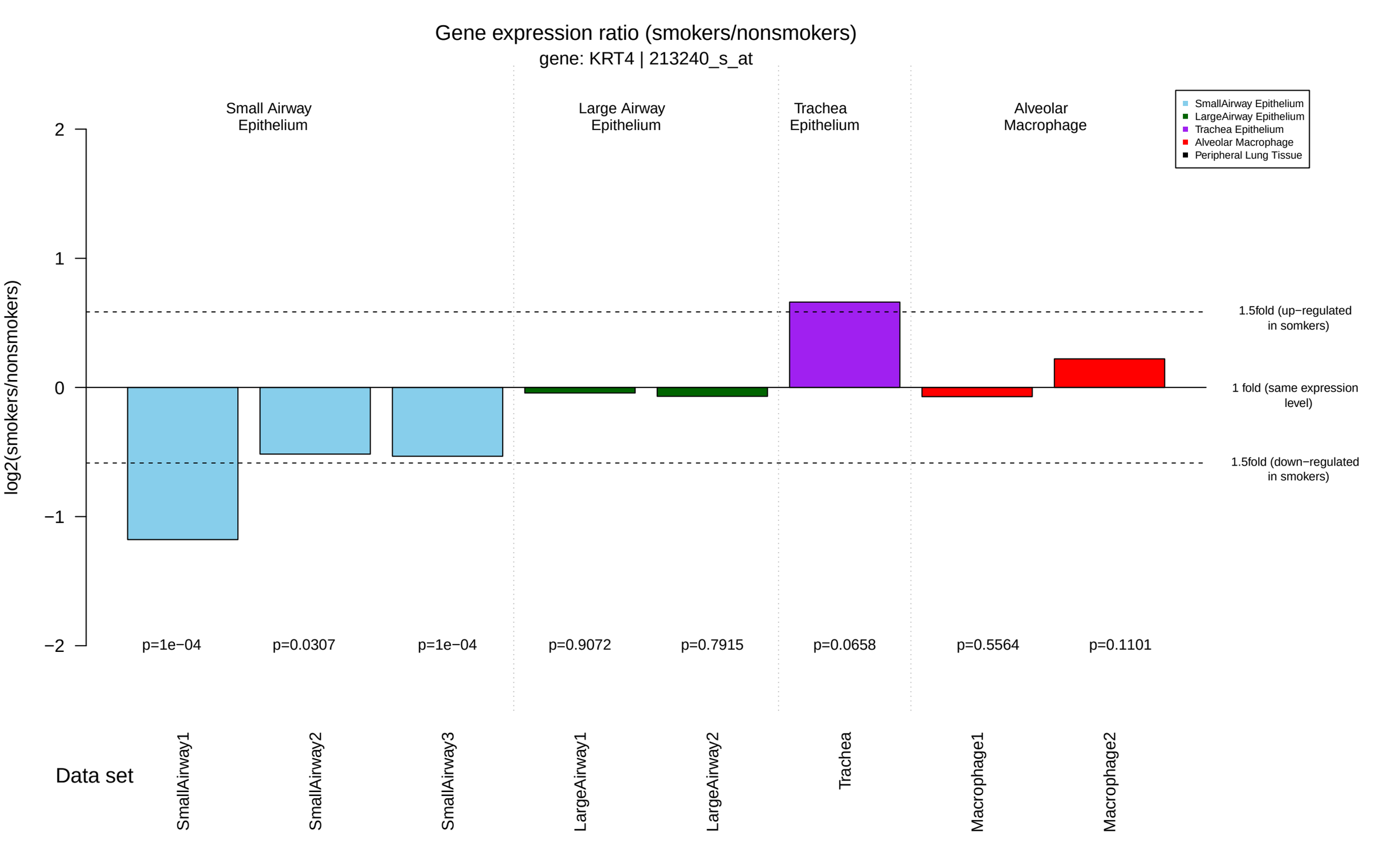
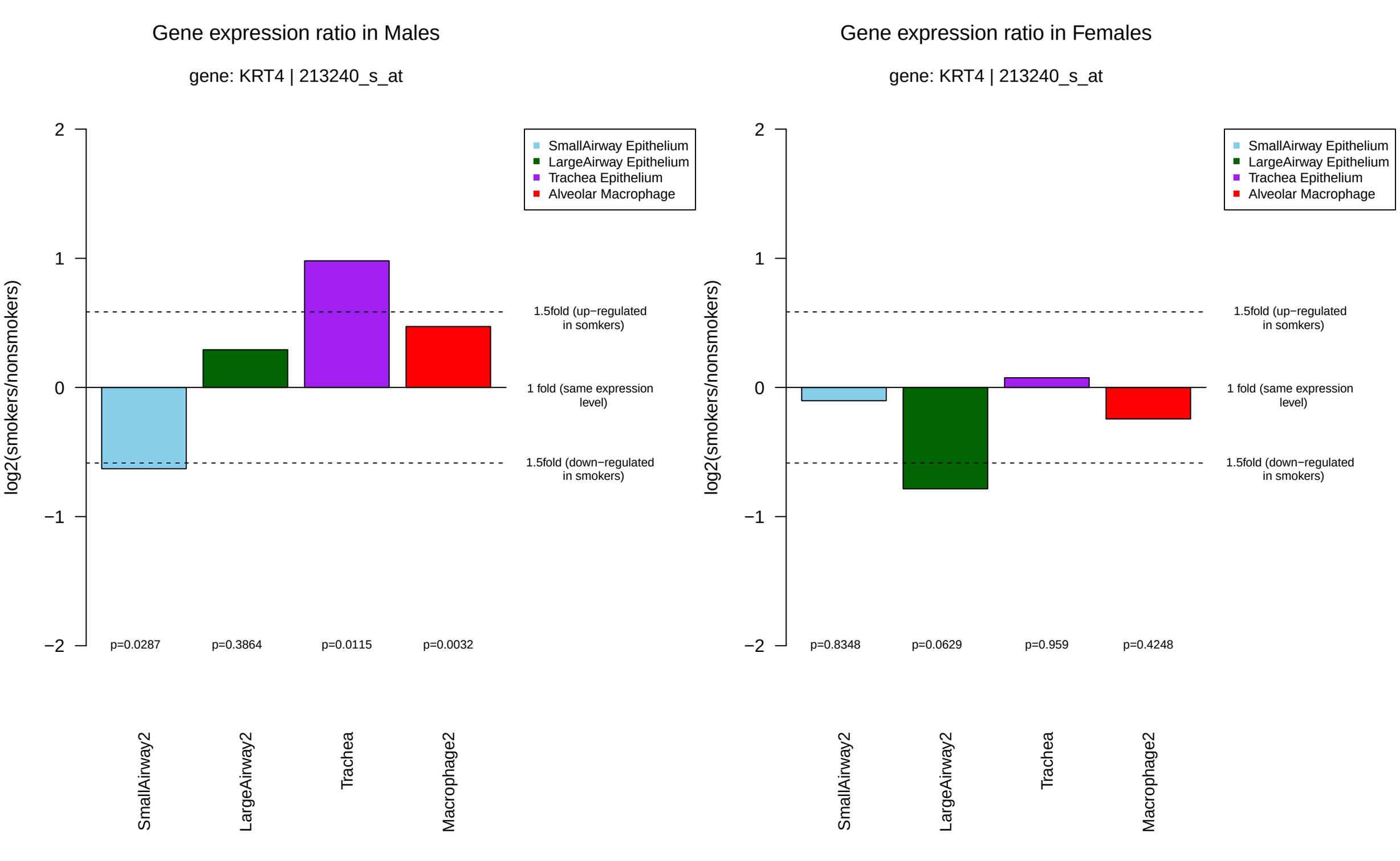
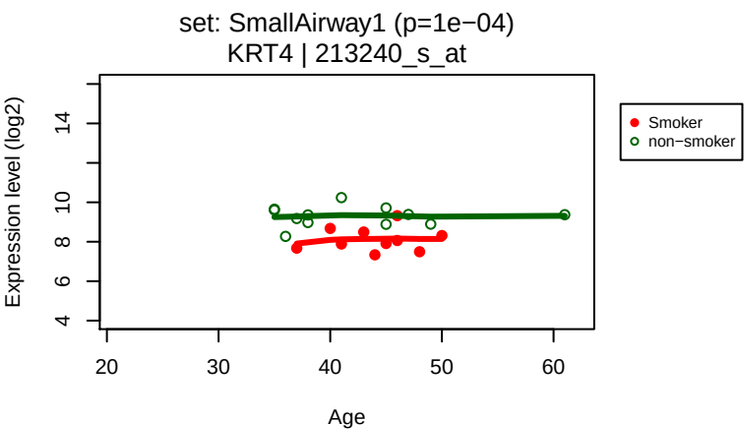
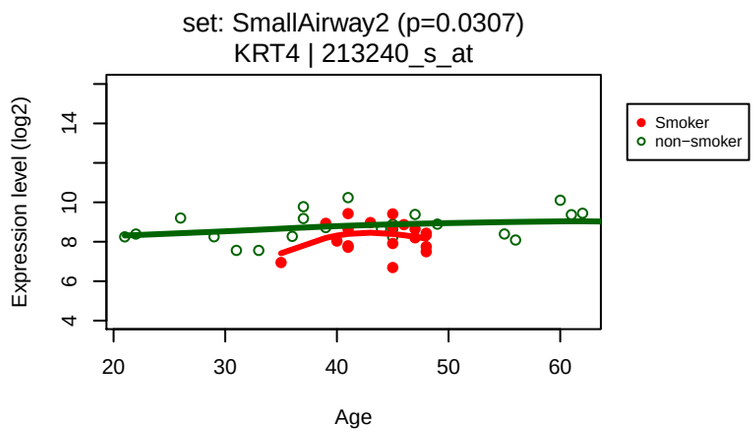
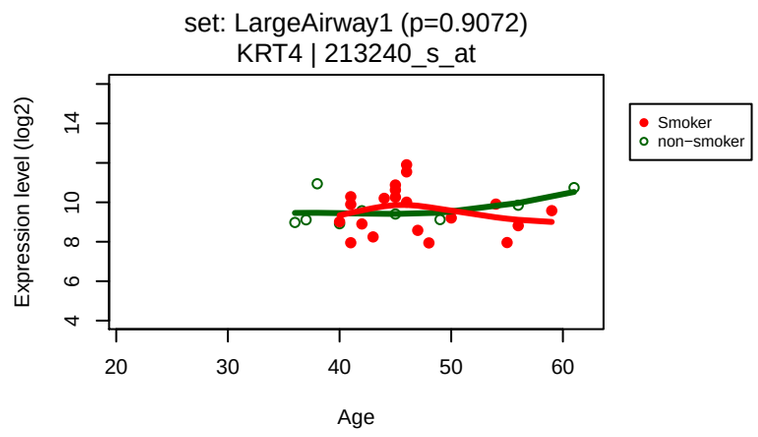

Gene expression of smokers and nonsmokers
gene: KRT4 | 213240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=1e−04
p=0.0307
p=1e−04
p=0.9072
p=0.7915
p=0.0658
p=0.5564
p=0.1101
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

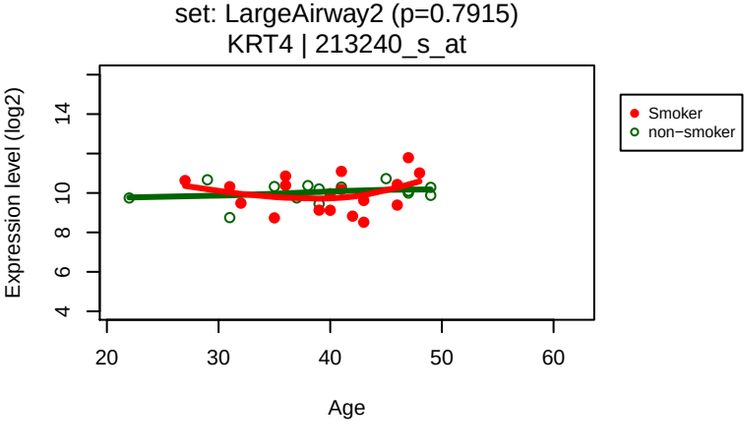

Gene expression ratio (smokers/nonsmokers)
gene: KRT4 | 213240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.0307
p=1e−04
p=0.9072
p=0.7915
p=0.0658
p=0.5564
p=0.1101
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

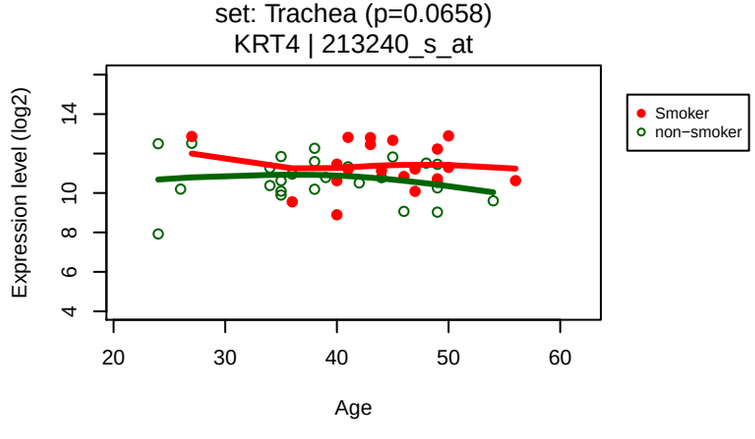

Gene expression ratio in Males
gene: KRT4 | 213240_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0287
p=0.3864
p=0.0115
p=0.0032
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRT4 | 213240_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8348
p=0.0629
p=0.959
p=0.4248
SmallAirway2
LargeAirway2
Trachea
Macrophage2

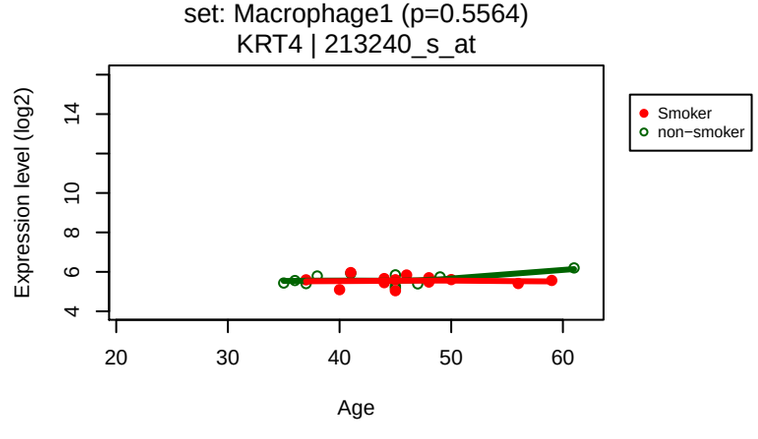

set: SmallAirway1 (p=1e−04)
KRT4 | 213240_s_at
Expression level (log2)
Age
Smoker
non−smoker

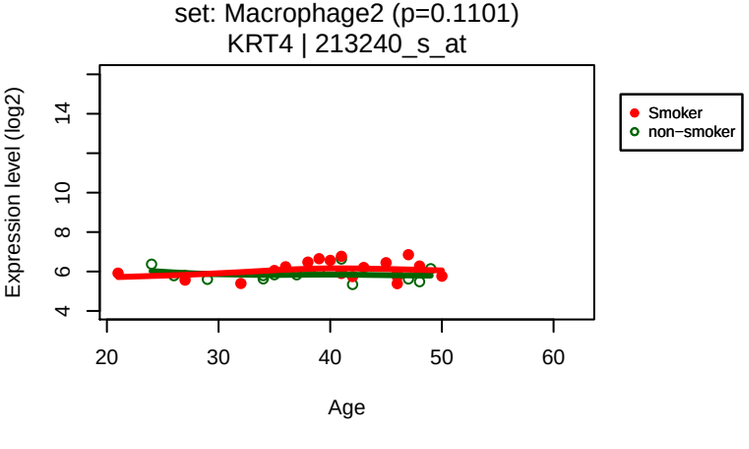

set: SmallAirway2 (p=0.0307)
KRT4 | 213240_s_at
Expression level (log2)
Age
Smoker
non−smoker

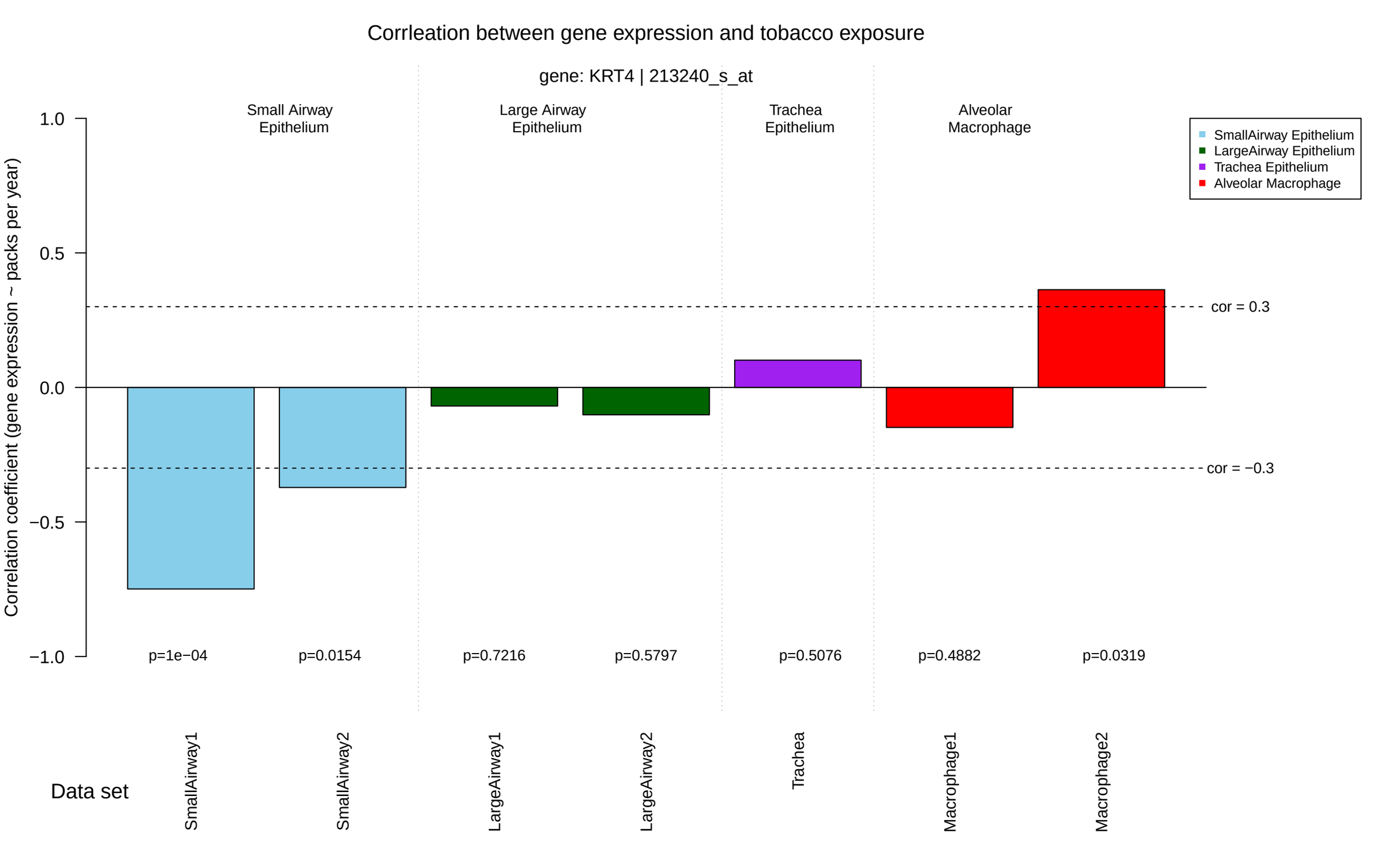

set: LargeAirway1 (p=0.9072)
KRT4 | 213240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7915)
KRT4 | 213240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0658)
KRT4 | 213240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5564)
KRT4 | 213240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1101)
KRT4 | 213240_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRT4 | 213240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=1e−04
p=0.0154
p=0.7216
p=0.5797
p=0.5076
p=0.4882
p=0.0319
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2